

ASHWIN SATISH

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EDUCATION

Northeastern University - Khoury College of Computer Science

Sept 2024 - Dec 2026 | Boston, MA

Master of Science, Data Science

GPA: 3.83/4.0

Machine Learning, NLP, MLOps, Algorithms, Data Mining

Christ University

June 2020 - May 2023 | Bangalore, India

Bachelor of Science, Computer Science

GPA: 3.53/4.0

EXPERIENCE

Gyori Lab for Computational Biomedicine | [Machine Learning Developer Intern](#)

Jan 2026 - Present | Boston, MA

- Engineered [TrialSynth](#)'s Trial Result LLM extraction pipeline in **Python** by parsing **35M** PubMed records, identifying **278K+** **publications** linked to registered trials, transforming **13K+** **papers** into structured data for clinical-trial knowledge graphs.
- Designed & built [CoGEx](#)'s Trial Result data model, transforming trial datasets to a **Neo4j graph**, generating **115K+** nodes & **116K+** relationships with **REST APIs** & [Flask search interface](#) enabling queryable clinical trials within the knowledge graph.
- Architected a custom grounding pipeline for Trial Result data by integrating the Ontology of Adverse Events (OAE), improving adverse events grounding coverage from **75.1% to 85.4%**.
- Benchmarked LLMs on **AWS Bedrock** for clinical-data extraction across cost, recall, and token efficiency, cutting per-paper inference cost **~65%** while matching **90% recall** against the baseline model.

Northeastern University | [Machine Learning Researcher](#)

Jan 2025 - Dec 2025 | Boston, MA

- Boosted true positives from **2% to 77%** (maintaining zero false positives) for Lyme cell detection via **7-model CNN ensemble**.
- Built a **Python** pipeline for reconstruction & masking of **30,000+** **cell images**, saving the biology team weeks of manual work.
- Improved accuracy by **13%** via augmentation & L2/dropout regularization, achieving production-ready performance.

Northeastern University | [Teaching Assistant \(Foundations of Data Science\)](#)

Aug 2025 - Dec 2025 | Boston, MA

- Mentored & conducted office hours for **300+** **students** in linear algebra & statistics via problem-solving & **NumPy** demos.

CarerAI | [Data Science Intern](#)

Jul 2023 - May 2024 | Gurgaon, India

- Prevented **2,000+** **unnecessary procedures** for customers by uncovering **500+** surgery avoidance cases through KNN analysis of patient records against historical data.
- Leveraged EHR for a **risk-score**, improving classification accuracy by **26.4%**, increasing confidence in surgery avoidance cases.
- Improved KNN's score by **20%** using DBSCAN to remove record outliers, reducing false positives in surgery recommendations.

SKILLS

Programming & Data: Python, SQL, R, Pandas, NumPy, PySpark, Scikit-learn, TensorFlow, PyTorch, Hugging Face, OpenCV

Cloud & Tools: Git, Tableau, Matplotlib, Seaborn, AWS, GCP, Azure, PostgreSQL, Snowflake, Docker, Neo4j, Bedrock, Lambda

AI & Modeling: Transformers, BERT, LoRA, AutoML, T5, LangChain, LangGraph, Kubeflow, Ollama, PydanticAI, MLFlow, Keras

PROJECTS

AI-Powered Pokémon Battle Strategy Engine | [Github](#)

- Built & dockerized hybrid **Multi-Agent System (MAS)** integrating **Llama 3.2** with deterministic backend for strategy analysis with a recommendation system for optimal team synergies by ingesting **1M+** aggregated usage **stats** via **Python ETL** pipeline.
- Engineered a RAG pipeline via ChromaDB & Sentence-Transformers, **indexing 40,000+** strategy docs to ground LLM inference in semantic context.

PantryPilot: Full-Stack Pantry & Recipe Assistant (2nd Place | MLOps Innovation Expo @ Google) | [Github](#)

- Contributed to the fine tuning, evaluation & automated deployment (CI/CD) on **GCP** of **Llama 3.2 3B** for personalized inventory recipes, resulting in **50% reduction** in dietary violations via LoRA adaptation.
- Collaborated on DPO feedback with Teacher model (**Llama 3 70B**) for preference labelling, VLM for OCR & Veo 3.1 for video.

Airlytics for Fitbit: Analysis, Forecast & Health Coach | [Github](#)

- Built personalized readiness-forecasting models analyzing Fitbit Air's live & historical data via Python ETL pipeline, **beating a naive baseline by 16%**.
- Built LLM health coach that provides health insights on retrieved data, validated by scoring tool-call accuracy & grounding rate.